



An algebraic investigation of Linear Logic

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Abstract

In this paper we investigate two logics (and their fragments) from an algebraic point of view. The two logics are: MALL (multiplicative-additive Linear Logic) and LL (classical Linear Logic). Both logics turn out to be strongly algebraizable in the sense of Blok and Pigozzi and their equivalent algebraic semantics are, respectively, the variety of Girard algebras and the variety of girales. We show that any variety of girales has a TD-term and hence equationally definable principal congruences. Also we investigate the structure of the algebras in question, thus obtaining a representation theorem for Girard algebras and girales. We also prove that congruence lattices of girales are really congruence lattices of Heyting algebras, thus determining the simple and subdirectly irreducible girales. Finally we introduce a class of examples showing that the variety of girales contains infinitely many nonisomorphic finite simple algebras.

Keywords Linear Logic · Algebraic semantics · Girales

Mathematics Subject Classification 03G27

1 Foreword

This note is a reworking of a manuscript (which dates back to 1996) that had a very limited circulation at the time and was never published. The aim of the research was to investigate Linear Logic and its various extensions and fragments from an algebraic point of view, making use of the (then) new results about algebraization by Blok and Pigozzi [11]. Algebraic models for Linear Logic were not a novelty even then: A. Avron proposed an algebraic semantics derived from an Hilbert-style system in [8]. In the second half of the 1990's A. Ursini wrote a long draft (titled *Semantical investigations of Linear Logic*) which was (sadly) never published and it is not even available online.

However at the time nobody had really tackled Linear Logic using the Blok–Pigozzi machinery and I proceeded to do exactly that. Unfortunately the paper was not received

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very favorably; it was made very clear to me that the interests of linear logicians focused more on computation and less on abstract provability and that the use of algebra was not appreciated among them. Not wanting to look like a novel Don Quixote I decided to drop the matter and put the paper in a (metaphorical) drawer where it sat for 27 years.

Of course things change with time; due to its enormous success, Abstract Algebraic Logic became more and more central among algebraists and logicians. And apparently the community of linear logicians has partly abandoned the strict orthodoxy of the beginning. In 2023 I was contacted by Wesley Fussner who was using some of my results in one of his projects. I was surprised that he knew about the manuscript and I checked: as bad as it was, the manuscript has managed to collect a handful of citations in Google Scholar. Wesley invited me to upload it on arXiv, so that he could properly cite it. I was reluctant to do that, but I dusted it a little bit and put it there against my best judgement. Later further conversations with him and other colleagues convinced me that, with some effort, a proper publication could be possible. So I subjected the arXiv version to a more vigorous dusting and this is the result.

For all the unexplained notions in universal algebra, we direct the reader to [14]; for the fine points of the Blok–Pigozzi algebraization procedure and related concepts our references are [11] and [19].

2 Preliminaries

When one wishes to investigate a (nonclassical) logic one has a choice between two approaches: the syntactical and the algebraic. The first usually gives rise to a *relational (Kripke-style) semantics*, while the other deals with *algebraic semantics*. The great success of Kripke in the sixties with his relational semantics for modal and intuitionistic logic was a source of inspiration for many researches based on his methods, while the algebraic approach receded into the background. The algebraic approach became fashionable again starting in the late seventies, mainly because of the work of W. Blok and D. Pigozzi. W. Blok, in his Ph.D. thesis [9], conducted an in-depth study of Lewis' modal logic S4, and in [10] he investigated the entire lattice of modal logics by purely algebraic means. Later he and D. Pigozzi investigated thoroughly the matter of algebraizability of logics [11, 12]. This investigation set the foundation for a new field, now commonly called *abstract algebraic logic*.

One of the first results of their line of investigation was the identification of the “right” concept of *algebraizable logic*. Roughly speaking, a logic L is algebraizable if there is a class K of structures which is to L what the variety of Boolean algebras is to classical propositional calculus. The class K is called the *equivalent algebraic semantics* of L .

If we understand a logic as a consequence operator, then its equivalent algebraic semantics (if any) consists of algebras (no infinitary operations, no relations, no second order axioms) and it is always at least a quasivariety. A logic that has an equivalent algebraic semantics is said to be **algebraizable**; if the equivalent algebraic semantics happens to be a variety, then the logic is **strongly algebraizable**.

Let ρ be an algebraic language; by $\mathbf{T}_\rho(\omega)$ we mean the algebra of terms of type ρ ; if $\mathbf{A}$ is an algebra of type ρ and p, q n -ry terms in $\mathbf{T}_\rho(\omega)$ we say that $\mathbf{A}$ is a **model of** $p \approx q$ (and we write $\mathbf{A} \models p \approx q$) if for all $\mathbf{a} \in A^n$, $p(\mathbf{a}) = q(\mathbf{a})$. If $\mathbf{K}$ is a class of algebras of type ρ , then we say that $\mathbf{K}$ is a **model of** $p \approx q$ if $\mathbf{A} \models p \approx q$ for all $\mathbf{A} \in \mathbf{K}$. In this case we will write $\models_{\mathbf{K}} p \approx q$ and $\models_{\mathbf{K}}$ is often called the **semantical consequence of** $\mathbf{K}$.

In loose terms, to establish the algebraizability of a logic $\mathbf{L}$ with respect to a quasivariety of algebras $\mathbf{Q}_{\mathbf{L}}$ over the same language ρ , one needs a finite set of one-variable equations (the **defining equations**)

$$\tau(x) = \{\delta_i(x) \approx \varepsilon_i(x) : i = 1, \dots, n\}$$

over terms of type ρ and a finite set of formulas (the **congruence formulas**) of $\mathbf{L}$ in two variables

$$\Delta(x, y) = \{\varphi_1(x, y), \dots, \varphi_m(x, y)\}$$

that allow to transform equations, quasiequations and clauses in $\mathbf{Q}_{\mathbf{L}}$ into formulas, single- and multiple-conclusion rules of $\mathbf{L}$; moreover this transformation must intuitively respect both the consequence relation of the logic and the semantical consequence of the quasivariety; more precisely, for all sets of formulas Γ of $\mathbf{L}$ and formulas $\varphi \in \mathbf{T}_\rho(\omega)$

$$\Gamma \vdash_{\mathbf{L}} \varphi \quad \text{iff} \quad \tau(\Gamma) \models_{\mathbf{Q}_{\mathbf{L}}} \tau(\varphi)$$

where $\tau(\Gamma)$ is a shorthand for $\{\tau(\gamma) : \gamma \in \Gamma\}$, and also

$$(x \approx y) \models_{\mathbf{Q}_{\mathbf{L}}} \tau(\Delta(x, y)).$$

We mention that algebraizability is preserved by extensions (and strong algebraizability by axiomatic extensions).

The knowledge that a given quasivariety $\mathbf{Q}$ of algebras is the equivalent algebraic semantics of a known logical system yields a good deal of information on its algebraic structure. Conversely, one can discover algebraic properties of members of $\mathbf{Q}$ that can be transformed into logical data.

In this note we will apply this machinery to two logics: *multiplicative-additive linear logic* MALL and *classical linear logic* LL. Both MALL and LL will turn out to be strongly algebraizable. As is usually the case it is no surprise what their equivalent classes are: they consist of residuated lattices (possibly with a modal operator) obeying equations reflecting the logical axioms.

3 Linear Logic

Linear Logic is a “resource-conscious” logic introduced by J.-Y. Girard in the late 80’s [20]. Since then Linear Logic has been developed by Girard himself, by his school and by many, many others the full list of whom would be too long to include here.

Linear Logic is resource conscious in that the left side of a sequent represents a resource that cannot be used freely. In a Gentzen-style axiomatization this consciousness shows itself by the absence of the classical weakening and contraction rules. For instance $A, A \vdash B$ means that we use two resources of type A to get a datum of type B . Moreover—and this is the main difference from other substructural logics—Girard introduced two operators (the *exponentials*) that serve to allow weakening and contraction in a controlled way on individual formulas.

The propositional language of Linear Logic consists of four families of connectives¹:

- The *multiplicative* connectives: $\cdot$, $\wp$ (the par, i.e. the parallel “or”), $\rightarrow$ (the linear implication, Girard’s $\multimap$), 0 and 1 ;
- the *additive* connectives: $\vee$, $\wedge$, $\top$ and $\perp$;
- The *linear negation* $\neg$, which is a de Morgan involution with respect to $\vee$ and $\wedge$;
- The *exponentials*: $!$ and $?$.

A suggestive way of thinking about how these connectives work is to view formulas as data types. For instance $A \wedge B$ is a datum from which we can extract, once, either a datum of type A and a datum of type B ; $A \cdot B$ is just a pair of data; $A \rightarrow B$ is a method of transforming a single datum of type A into a datum of type B ; $!A$ indicates that we can extract as many data of type A as we like (weakening and contraction on the left side of a sequent); and so on.

The original formulation of Linear Logic is in a Gentzen-style axiomatization. However, in order to take advantage of Blok–Pigozzi’s theory of algebraizability, it is helpful to look at Linear Logic as a 1-deductive system in the sense of [12].

In his original paper [20], J.-Y. Girard gave absolutely no meaning to the concept of “linear logical theory” or to any kind of associated consequence relation. The question was tackled again by A. Avron [8]. He observed that the classical methods for associating a consequence relation to the Gentzen-type presentation of Linear Logic gives rise to two meaningful consequence relations.

- (The *internal* consequence relation)

$$\varphi_1, \dots, \varphi_n \vdash_{\text{LL}}^I \psi$$

iff the corresponding sequent is derived in the Gentzen-type formalism iff $\varphi_1 \rightarrow (\varphi_2 \rightarrow (\dots (\varphi_n \rightarrow \psi) \dots))$ is a theorem of Linear Logic.

- (The *external* consequence relation)

$$\varphi_1, \dots, \varphi_n \vdash_{\text{LL}}^E \psi$$

¹ Our notation is slightly different from the original formulation, in that 0 and $\perp$ are exchanged; the reason is that the original formulation conflicts with the common usage in residuated lattices.

iff the sequent $\Rightarrow \psi$ is derivable in the Gentzen-type formalism obtained from the linear one by adding $\Rightarrow \varphi_1, \dots \Rightarrow \varphi_n$ as axioms.

He also gave an Hilbert-style presentation of LL with associated consequence relation $\vdash_{\text{LL}}$ and (Theorem 2.7 in [8]) showed that² $\Sigma \vdash_{\text{LL}} \varphi$ if and only if $\Sigma \vdash_{\text{LL}}^E \varphi$; this seems to imply that the view of Linear Logic as a deductive system is not totally disconnected from its motivations.

Here is the Hilbert-style axiomatization of LL as presented by A. Avron [8], consisting of twenty-four axioms and three inference rules.

(HL1) $p \rightarrow p$	(HL2) $(p \rightarrow q) \rightarrow ((q \rightarrow r) \rightarrow (p \rightarrow r))$
(HL3) $(p \rightarrow (q \rightarrow r)) \rightarrow (q \rightarrow (p \rightarrow r))$	(HL4) $\neg\neg p \rightarrow p$
(HL5) $(p \rightarrow \neg q) \rightarrow (q \rightarrow \neg p)$	(HL6) $p \rightarrow (q \rightarrow p \cdot q)$
(HL7) $p \rightarrow (q \rightarrow r) \rightarrow (p \cdot q \rightarrow r)$	(HL8) 1
(HL9) $1 \rightarrow (p \rightarrow p)$	(HL10) $p \rightarrow (\neg p \rightarrow 0)$
(HL11) $\neg 0$	(HL12) $p \wedge q \rightarrow p$
(HL13) $p \wedge q \rightarrow q$	(HL14) $(p \rightarrow q) \wedge (p \rightarrow r) \rightarrow (p \rightarrow q \wedge r)$
(HL15) $p \rightarrow p \vee q$	(HL16) $q \rightarrow p \vee q$
(HL17) $(p \rightarrow r) \wedge (q \rightarrow r) \rightarrow (p \vee q \rightarrow r)$	(HL18) $p \rightarrow \top$
(HL19) $\perp \rightarrow p$	(HL20) $q \rightarrow (!p \rightarrow q)$
(HL21) $(!p \rightarrow (!p \rightarrow q)) \rightarrow (!p \rightarrow q)$	(HL22) $!(p \rightarrow q) \rightarrow (!p \rightarrow !q)$
(HL23) $!p \rightarrow p$	(HL24) $!p \rightarrow !!p$
(MP) $\frac{p \quad p \rightarrow q}{q}$	(Adj) $\frac{p \quad q}{p \wedge q}$
(Nec) $\frac{p}{!p}$	

For the logical reasons why we need to introduce the (Adj) rule, we refer the reader to [8], p.171. Moreover:

- In LL one can define $p \wp q$ as $\neg p \rightarrow q$.
- In LL one can define $?p$ as $\neg!(\neg p)$.
- Finally $\vdash_{\text{MALL}} p \rightarrow q$ if and only if $\vdash_{\text{LL}} \neg(p \cdot \neg q)$.

4 The algebraization

While the exponentials $!$ and $?$ were central in Girard's original idea, the *exponential-free fragment* of Linear Logic has also attracted a lot of interest. If T is a subset of the set of connectives of LL we define the T -fragment of LL in the following way. The language is obtained by deleting all the connectives not belonging to T ; the valid derivations of the T -fragment are the valid derivations in LL in which only connective from T appear. Similarly the T -free fragment of LL contains only connectives not in T and as valid derivations those that contain only connectives not in T .

Then MALL, the *multiplicative-additive linear logic*, is the exponential-free fragment of LL. It is clearly axiomatized by (HL1)-(HL19) plus (MP) and (Adj). Let us stress that MALL is a honest-to-God substructural logic close to the well-studied system

² He really proved it for MALL but the proof carries through easily for LL.

R of *Relevance Logic* [7]. This connection can be roughly expressed by the equation

$$R - \text{contraction} = \text{MALL} + \text{distribution}.$$

Techniques from R have been applied to MALL with some success (see for instance [6]). Moreover MALL is superior to R at least in that it has a *cut-elimination* theorem. However, everything comes with a price tag: the lack of distribution in MALL makes things harder from an algebraic point of view.

The relationships between MALL and LL is more complex than it seems. We observe that the axioms (HL22)–(HL24) and (Nec) mimic the introduction of a S_4 modality; the idea of viewing LL as a modal logic, which is close to hereticism for linear logicians, is getting more and more traction lately.

We observe also that (HL20) and (HL21) (weakening and contraction limited to exponentiated formulas) seem to be responsible for a quantum leap: MALL is decidable (really, *PSPACE-complete*) while LL is not [23].

Here is the main algebrization result.

Theorem 4.1 *Any fragment of MALL containing $\{\wedge, \rightarrow, 1\}$ is algebraizable with defining equation $p \wedge 1 = 1$ and congruence formulas $\Delta(p, q) = \{p \rightarrow q, q \rightarrow p\}$.*

Proof Let's write from now on $p \Delta q$ for $\Delta(p, q)$. Then the derivations

$$\begin{aligned} &\vdash p \Delta p \\ &p \Delta q \vdash q \Delta p \\ &p \Delta q, q \Delta r \vdash p \Delta r \end{aligned}$$

follow readily from axioms (HL1)–(HL3) and preservation by connectives is easily checked.

It remains to show that $p \wedge 1 \Delta 1 \vdash p$ and that $p \vdash p \wedge 1 \Delta 1$. Thus for the first

$$\begin{array}{ll} 1 \rightarrow p \wedge 1 & \text{(because of the defining equation)} \\ 1 & \text{(HL8)} \\ p \wedge 1 & \text{(MP)} \\ p & \text{(HL12)} \end{array}$$

and for the second

$$\begin{array}{ll} 1 \rightarrow (p \rightarrow p) & \text{(HL9)} \\ p \rightarrow (1 \rightarrow p) & \text{(HL3) + (MP)} \\ p & \text{(Hyp)} \\ 1 \rightarrow p & \text{(MP)} \\ 1 \rightarrow 1 & \text{(HL1)} \\ (1 \rightarrow p) \wedge (1 \rightarrow 1) \rightarrow (1 \rightarrow p \wedge 1) & \text{(HL14)} \\ 1 \rightarrow (p \wedge 1) & \text{(MP)} \\ p \wedge 1 \rightarrow 1 & \text{(HL13)} \\ p \wedge 1 \Delta 1 & \end{array}$$

□

Let T be any fragment of MALL containing $\{\rightarrow, \wedge, 1\}$. The type of its equivalent algebraic semantics Q_T is determined by the connectives in T and we will follow the common usage of denoting them by the same symbols. Moreover

$$\Gamma \vdash_T p \quad \text{iff} \quad \{q \wedge 1 = 1 : q \in \Gamma\} \models_{Q_T} p \wedge 1 = 1.$$

A (pointed) **Girard semilattice** is an algebra $\langle A, \rightarrow, \wedge, 1 \rangle$ where, $\langle A, \wedge, 1 \rangle$ is a pointed semilattice and moreover for all $a, b, c \in A$

$$1 \rightarrow a = a \tag{L1}$$

$$a \rightarrow a \geq 1 \tag{L2}$$

$$(a \rightarrow b) \wedge (a \rightarrow c) = a \rightarrow (b \wedge c) \tag{L3}$$

$$a \rightarrow b \leq (c \rightarrow a) \rightarrow (c \rightarrow b) \tag{L4}$$

$$a \rightarrow (b \rightarrow c) \leq b \rightarrow (a \rightarrow c) \tag{L5}$$

$$a \rightarrow b, b \rightarrow a \geq 1 \quad \text{implies} \quad a = b \tag{L6}$$

Girard semilattices form a quasivariety GS that is not a variety (this can be shown by an easy reworking of known examples). Let us also observe that $\rightarrow$ is a BCI implication that is not a residuation (so a Girard semilattice is not in general a residuated semilattice in the sense of [25]).

Theorem 4.2 *If $T = \{\rightarrow, \wedge, 1\}$ then Q_T is the quasivariety of Girard semilattices.*

Proof Let $\mathbf{A} \in Q_T$. We define a relation on A by setting

$$a \leq b \quad \text{iff} \quad (a \rightarrow b) \wedge 1 = 1.$$

The relation is reflexive by (H1), transitive by (H2) and the congruence formulas imply antisymmetry. Hence $\leq$ is a partial order on A and (H12–H14) imply that $a \wedge b$ is the greatest lower bound of a and b , making $\mathbf{A}$ a pointed semilattice in which (L6) holds.

The rest consists of standard calculations; first observe that (L2) is a direct consequence of (HL9), (L3) comes in the same fashion from (HL14), (L4) from (HL2) and (HL3) and (L5) from (HL3). For (L1) we observe that the following derivations hold in T :

$$\begin{aligned} 1 &\rightarrow (p \rightarrow p) \text{ (HL9)} \\ p &\rightarrow (1 \rightarrow p) \text{ (HL3) + (MP)} \end{aligned}$$

$$\begin{aligned}
(1 \rightarrow p) &\rightarrow (1 \rightarrow p) \text{ (HL1)} \\
1 &\rightarrow ((1 \rightarrow p) \rightarrow p) \text{ (HL3) + (MP)} \\
(1 \rightarrow p) &\rightarrow p \quad \quad \quad \text{(HL8) + (MP)}
\end{aligned}$$

Via the usual translation this implies (L1). $\square$

Introducing the join causes no problems using (HL17); an algebra $\langle A, \rightarrow, \vee, \wedge, 1 \rangle$ is a **Girard lattice** if

- $\langle A, \vee, \wedge \rangle$ is a lattice;
- $\langle A, \rightarrow, \wedge, 1 \rangle$ is a Girard semilattice;
- for all $a, b, c \in A$

$$(a \rightarrow c) \wedge (b \rightarrow c) = (a \vee b) \rightarrow c. \quad (\text{L7})$$

Clearly the equivalent algebraic semantics of the $\{\rightarrow, \vee, \wedge, 1\}$ -fragment is the quasi-variety GL of Girard lattices.

Introducing the binary multiplicative connective can be done in a standard way: thanks to (HL6) and (HL7), $(\rightarrow, \cdot)$ form a residuated pair. This implies that the equivalent algebraic semantics of the $\{\wedge, \rightarrow, \cdot, 1\}$ -fragment is the variety of commutative residuated semilattices. Similarly the equivalent algebraic semantics of the $\{\vee, \wedge, \rightarrow, \cdot, 1\}$ -fragment is the variety CRL of **commutative residuated lattices**. With a slight abuse of language we will call this fragment the **positive fragment** of MALL and we will denote it by MALL^+ .

5 The first embedding

In this section we would like to show that MALL^+ is not a conservative extension of the $\{\rightarrow, \wedge, 1\}$ -fragment. The translation into algebraic terms consists in proving that the $\{\rightarrow, \wedge, 1\}$ -subreducts of algebras in CRL form a proper subclass (as a matter of fact a subvariety) of GS. This is a consequence of the following well-known fact observed in [3] and rediscovered many times in the literature.

Theorem 5.1 [3] *For every variety $\mathbf{V}$ of commutative residuated lattices, the class of $\{\wedge, \rightarrow, 1\}$ -subreducts of $\mathbf{V}$ is a variety.*

It is very easy to find a quasiequation holding in the varieties of $\{\rightarrow, \wedge, 1\}$ -subreducts but not in GS; in fact in CRL, $\rightarrow$ is a residuation and this somehow carries over in the sense that for all $\{\rightarrow, \wedge, 1\}$ -subreduct $\mathbf{A}$ and for all $a, b \in A$

$$a \rightarrow b \geq 1 \quad \text{implies} \quad a \leq b.$$

Now it is easily checked that this quasiequation does not hold in GS (since a Girard semilattice is not in general a residuated semilattice).

Let $\mathbf{V}$ be the variety of Girard semilattices satisfying the further equation

$$x \leq ((x \rightarrow y) \wedge 1) \rightarrow y \quad (\text{L8})$$

We claim that $\mathbf{V}$ is the variety of $\{\rightarrow, \wedge, 1\}$ -subreducts of CRL. Now it is easy to show that (L8) holds in any $\{\rightarrow, \wedge, 1\}$ -subreduct and implies (L6). Hence we only need to show that any member of $\mathbf{V}$ is embeddable in a commutative residuated lattice. First let's prove that the algebras in $\mathbf{V}$ have *residuals without residuations*:

Lemma 5.2 *Let $\mathbf{A} \in \mathbf{V}$ and let $a, b, c \in A$; then*

1. $a \leq b$ if and only if $a \rightarrow b \geq 1$;
2. $a \leq (a \rightarrow b) \rightarrow b$;
3. $a \leq b$ implies $b \rightarrow c \leq a \rightarrow c$ and $c \rightarrow a \leq c \rightarrow b$.

Proof Suppose $a \leq b$; then $a \wedge b = a$. Then by (L3)

$$(a \rightarrow a) \wedge (a \rightarrow b) = a \rightarrow (a \wedge b) = a \rightarrow a;$$

so $a \rightarrow b \geq a \rightarrow a \geq 1$ by (L2).

Conversely, assume $a \rightarrow b \geq 1$; then by (L8) and (L1)

$$a \leq ((a \rightarrow b) \wedge 1) \rightarrow b = 1 \rightarrow b = b.$$

For (2), by (L5) and (L2) we get

$$a \rightarrow ((a \rightarrow b) \rightarrow b) = (a \rightarrow b) \rightarrow (a \rightarrow b) \geq 1$$

and by (1) $a \leq (a \rightarrow b) \rightarrow b$.

The proof of (3) is routine using (1), (L3) and (L5). $\square$

The embedding we are going use is based on the theory of frames developed in [17]. Let $\mathbf{A} \in \mathbf{V}$ and let $\Gamma_{\mathbf{A}}$ be the set of semilattice filters of $\mathbf{A}$; we say that a subset $X \subseteq \Gamma_{\mathbf{A}}$ is **hereditary** if for all $F, G \in \Gamma_{\mathbf{A}}$, $F \in X$ and $F \subseteq G$ implies $G \in X$. We also define for $a \in A$, $\mathbf{a} = \{F \in \Gamma_{\mathbf{A}} : a \in F\}$ and we note that $\mathbf{a}$ is hereditary. Note that the intersection of any family of hereditary subsets is hereditary; so we can define a closure operator in which the closed subsets are precisely the hereditary subsets of $\Gamma_{\mathbf{A}}$. It follows that the hereditary subsets of $\Gamma_{\mathbf{A}}$ form an algebraic lattice $D(\mathbf{A})$ ordered by inclusion.

Next we define a ternary relation on $\Gamma_{\mathbf{A}}$; for $F, G, H \in \Gamma_{\mathbf{A}}$

$R(F, G, H)$ if and only if for all $a, b \in A$, $a \in F$ and $a \rightarrow b \in G$ implies $b \in H$.

This relation allows us to introduce additional operations: if $X, Y \in D(\mathbf{A})$

$$X \circ Y = \{H : \exists F \in Y, \exists G \in X \text{ with } R(F, G, H)\}$$

$$X \rightarrow Y = \{H : \forall F, G \text{ if } R(F, H, G) \text{ and } F \in X, \text{ then } G \in Y\}.$$

Of course the relationships between equations satisfied in $D(\mathbf{A})$ and the properties of R are relevant. In [16] there is a long list of these correspondences with no proofs, simply quoting the work of R. Routley and R. Meyer on the semantics of entailment

[26]; some proofs are indeed there, but they are so embedded in the general abstract theory of entailment that their connection to this algebraic setting is not immediately clear. That's why here we prefer to present direct proofs.

Lemma 5.3 *For any $A \in \mathbf{V}$, $\langle D(A), \circ, 1 \rangle$ is a commutative monoid.*

Proof Proving that $D(\mathbf{A})$ is closed under $\circ$ is straightforward. Let then $X, Y \in D(\mathbf{A})$ and suppose that $H \in X \circ Y$; then there is an $F \in Y$ and a $G \in F$ with $R(F, G, H)$. Let $a \in G$ and $a \rightarrow b \in F$; then (by Lemma 5.2 $a \leq (a \rightarrow b) \rightarrow b \in G$. So since $a \rightarrow b \in F$ and $R(F, G, H)$ we get $b \in H$; so $R(G, F, H)$ holds and hence $H \in Y \circ X$. This shows that $\circ$ is commutative.

Let ∇_A be the *positive cone* of $\mathbf{A}$, i.e. the principal filter generated by 1. Note that for any $F \in \Gamma_A$, $R(F, \nabla_A, F)$ holds and, since $\nabla_A \in 1$, we get at once that $X \subseteq 1 \circ X$. Conversely, let $H \in 1 \circ X$; then there is an $F \in X$ and a $G \in 1$ with $R(F, G, H)$. If $a \in F$, then $a \rightarrow a \geq 1 \in G$ and hence $a \in H$, so that $F \subseteq H$. But X is hereditary and $F \in X$, so $H \in X$ and eventually $1 \circ X = X$.

Associativity requires more work. Let $X, Y, Z \in D(\mathbf{A})$ with $H \in (X \circ Y) \circ Z$; then there is an $F \in Z$ and a $U \in X \circ Y$ with $R(F, U, H)$ and a $K \in X$ and a $G \in Y$ with $R(G, K, U)$. Let $L = \{d \in A : b \leq a \rightarrow d \text{ for some } a \in F, b \in G\}$; then using Lemma 5.2 we can show that $L \in \Gamma_A$ and clearly $R(F, G, L)$. Assume now $d \in L$ and $d \rightarrow c \in K$; again by Lemma 5.2, (L4) and (L5) we get

$$d \rightarrow c \leq (a \rightarrow d) \rightarrow (a \rightarrow c)$$

so $(a \rightarrow d) \rightarrow (a \rightarrow c) \in K$. Since $d \in L$ there are $a \in F$ and $b \in G$ with $b \leq a \rightarrow d$, so $a \rightarrow d \in G$ and, since $R(G, K, U)$, we get $a \rightarrow c \in U$. But $a \in F$, $a \rightarrow c \in U$ and $R(F, U, H)$ implies $c \in H$. Hence we conclude that $R(L, K, H)$.

Now by definition $L \in Y \circ Z$ and hence, since $R(F, G, L)$, $H \in X \circ (Y \circ Z)$; we have thus proved that $(X \circ Y) \circ Z \subseteq X \circ (Y \circ Z)$. The opposite inclusion follows from a similar argument, hence $\circ$ is associative. $\square$

Lemma 5.4 *For each $A \in \mathbf{V}$, $(\rightarrow, \circ)$ form a residuated pair w.r.t. the lattice ordering of $D(\mathbf{A})$.*

Proof Since we already know that $D(\mathbf{A})$ is closed under $\circ$ we have only to check that it is closed under $\rightarrow$ as well. Let then $H \in X \rightarrow Y$; then if $H \subseteq H'$ and $a \rightarrow b \in H'$, then $a \rightarrow b \in H$. It follows that if $R(F, H', G)$, then $R(F, H, G)$ for all $F, G \in \Gamma_A$ and so $H' \in X \rightarrow Y$ which is then hereditary.

Next we have to show that

$$X \circ Y \subseteq Z \quad \text{if and only if} \quad X \subseteq Y \rightarrow Z.$$

Assume then that $X \circ Y \subseteq Z$ and let $H \in X$. Let $F \in Y$ such that $R(F, H, G)$; then by definition $G \in X \circ Y$ and hence $G \in Z$. But this implies $H \in Y \rightarrow Z$, as wished. Conversely suppose $X \subseteq Y \rightarrow Z$ and let $H \in X \circ Y$; then there are $F \in Y$ and $G \in X$ with $R(F, G, H)$. But then $G \subseteq Y \rightarrow Z$, so if $F \in Y$ and $R(F, G, H)$, then $H \in Z$ as wished. $\square$

Hence we have shown that:

Theorem 5.5 *For any $A \in \mathbf{V}$, $D(A) = \langle D(A), \rightarrow, \vee, \wedge, \circ, 1 \rangle$ is commutative residuated lattice.*

Finally we prove the embedding.

Theorem 5.6 *Any algebra $A \in \mathbf{V}$ is embeddable in $D(A)$.*

Proof Define a mapping $h : A \mapsto H(A)$ by

$$h(a) = \mathbf{a}.$$

We start showing that for any $a, b \in A$, $H \in h(a \rightarrow b)$ if and only if $H \in \mathbf{a} \rightarrow \mathbf{b}$. This is equivalent to showing that, for $a, b \in A$

$$a \rightarrow b \in H \quad \text{if and only if} \quad \forall F, G \text{ if } R(F, H, G) \text{ and } a \in F, \text{ then } b \in G.$$

The left-to-right implication is a straightforward consequence of the definitions. Assume now that $a \rightarrow b \notin H$; we will show that there exists F, G with $a \in F$, $R(F, H, G)$ but $b \notin G$. Let's denote by $[a]$ the principal filter generated by a . Let $F = [a]$ and $G = \{d : \text{there is a } c \in H \text{ } c \leq a \rightarrow d\}$; note that $b \notin G$ otherwise $c \leq a \rightarrow b$ for some $c \in H$ and since H is a filter we would have $a \rightarrow b \in H$, contrary to the hypothesis. Again we can show that G is a filter using Lemma 5.2; so $G \in \Gamma_A$, $a \in F$ and $b \notin G$. Now we show that $R(F, H, G)$; if $u \in F$ and $u \rightarrow v \in H$, then $a \leq u$ and hence $u \rightarrow v \leq a \rightarrow v$, which by definition implies $v \in G$ and thus $R(F, H, G)$.

Now $a \leq b$ implies that $\mathbf{a} \subseteq \mathbf{b}$, so h is order preserving; hence to conclude the proof it is enough to show that h is injective. But if $\mathbf{a} = \mathbf{b}$, since $[a] \in \mathbf{a}$, we get $[a] \in \mathbf{b}$ so $b \in [a]$; by the same fashion $a \in [b]$ and hence $a = b$. $\square$

6 Algebraizing MALL and LL

For the complete MALL we have to work a little bit more. An element a of a commutative residuated lattice is **involutive** if for all $b \in A$, $(b \rightarrow a) \rightarrow a = b$; it is well known [15] that if in case we define $\sim b := (b \rightarrow a) \rightarrow a$, then $\sim$ is a “negation” with the following properties

- $\sim \sim a = a$ (involutive);
- $\sim(a \vee b) = \sim a \wedge \sim b$ and $\sim(a \wedge b) = \sim a \vee \sim b$ (De Morgan);
- $a \leq b$ implies $\sim b \leq \sim a$ (antitonic);
- $\sim(a \cdot \sim b) = a \rightarrow b$ (contraposition).

It is easily seen that these properties are not independent; for instance any negation that is involutive and satisfies one of the other three properties must satisfy them all.

A structure $\langle A, \vee, \wedge, \rightarrow, \cdot, 0, 1 \rangle$ is a **Girard algebra** if

- $\langle A, \vee, \wedge, \rightarrow, \cdot, 1 \rangle$ is a commutative residuated lattice;

- 0 is an involutive element.

In this case $\sim x = x \rightarrow 0$ is an involutive and antitonic so it also De Morgan and satisfies contraposition; so $\rightarrow$ and $\cdot$ are definable in terms of each other. Conversely any commutative residuated lattice with a negation $\sim$ that is involutive and De Morgan can be seen as a Girard algebra upon defining $0 = \sim 1$. A **bounded Girard algebra** is a Girard algebra with an additional constant $\top$ satisfying $x \rightarrow \top \geq 1$; we define $\perp := \sim \top$. By the usual standard arguments we get:

Theorem 6.1 *The equivalent algebraic semantics of MALL is the variety of bounded Girard algebras.*

Note that there 0 and 1 can be in any ordering relation; in particular it may happen that $1 \leq 0$, which implies that MALL is not an explosive logic i.e. it is a paraconsistent logic. It is worth noting that explosivity in our case means that

$$0 = 0 \cdot 1 = 0 \cdot \sim 0 \leq a$$

for all a . This implies $0 = \perp$ and hence $1 = \top$; hence the minimal explosive extension of MALL has as equivalent algebraic semantics the variety of integral Girard algebras. Since in this case the negation is a orthocomplementation it can be also seen as the variety of (bounded) residuated ortholattices.

If we look at LL it is clear that adjoining the exponentials corresponds to considering Girard algebras superimposed with a certain S4 modality. A **girale** is an algebra $\langle A, \vee, \wedge, \rightarrow, \cdot, \sim, 1, ! \rangle$ where

- $\langle A, \vee, \wedge, \rightarrow, \cdot, 1 \rangle$ is a Girard algebra;
- $!$ is unary and for all $a, b \in A$

$$!1 = 1 \tag{G1}$$

$$!a \leq a \wedge 1 \tag{G2}$$

$$!a!b = !(a \wedge b) \tag{G3}$$

$$!!a = !a. \tag{G4}$$

Let's prove some algebraic properties of girales.

Lemma 6.2 *Let A be a girale; then for any $a, b, c \in A$*

1. $a \leq b$ implies $!a \leq !b$;
2. $b \leq !a \rightarrow b$;
3. $!a = !a!a$;
4. $ab \leq c$ implies $!a!b \leq !c$;
5. $a \geq 1$ implies $!a = 1$;
6. $!(!a!b) = !a!b \leq !(ab)$;
7. $!(a \rightarrow b) \leq !a \rightarrow !b$;

$$8. !a \rightarrow (!a \rightarrow b) \leq !a \rightarrow b.$$

Proof (1) is immediate from (G2) and (G3); next note that in any commutative residuated lattice $b \leq (a \wedge 1) \rightarrow b$, thus (2) follows from (G2), while (3) is again a straightforward consequence of (G3).

If $ab \leq c$, then $!(a \wedge b) = !a!b \leq c$; hence (4) follows from (1) and (G4) while (5) follows from (1), (G1) and (G2). For (6) we compute

$$!(!a!b) = !(a \wedge b) = !(a \wedge b) = !a!b.$$

Moreover $!a!b \leq ab$ so by (1) $!(!a!b) \leq !(ab)$ and hence $!a!b \leq !(ab)$. Next since $!a \leq a$, we get $a \rightarrow b \leq !a \rightarrow b$; so $(a \rightarrow b)!a \leq b$, so $!(a \rightarrow b)!a \leq b$ and by (6) $!(a \rightarrow b)!a \leq !b$. So (7) holds.

For (8) we observe that in any residuated lattice, if a is an idempotent element in a residuated lattice then for all b

$$a(a \rightarrow (a \rightarrow b)) = a^2(a \rightarrow (a \rightarrow b)) \leq b$$

and by residuation

$$a \rightarrow (a \rightarrow b) \leq a \rightarrow b.$$

Since $!a$ is idempotent by (3) we conclude that

$$!a \rightarrow (!a \rightarrow b) \leq (!a \rightarrow b)$$

so (8) follows. $\square$

Now using Lemma 6.2 and the usual techniques of algebraization of logical systems it is straightforward to show that:

Theorem 6.3 *LL is strongly algebraizable and its equivalent algebraic semantics is the variety $\mathbf{G}$ of bounded girales.*

So in a way girales have the same relationship to MALL that interior algebras have to classical logic; so they belong to the very general class of (residuated) lattices with a superimposed modality (more on that later).

7 The second embedding

We will show that LL is a conservative extension of MALL. Of course we will do it from the algebraic side, i.e. we will prove that the class of subreducts of girales to the type of Girard algebras is the variety of Girard algebras. This is equivalent to showing that any Girard algebra is embeddable in a girale; in order to do so we will collect several information, that will be useful for other investigations as well, on the algebraic structures of girales.

Let $\mathbf{P}$ be any poset; we say that Q is a **relatively complete** subset of P if for all $p \in P$

$$\sup\{q \in Q : q \leq p\} \quad \inf\{q \in Q : q \leq p\}$$

both exist.

Let $\mathbf{A}$ be a Girard algebra and let as usual $A^- = \{a : a \leq 1\}$; a **relatively complete Heyting** subset of A is a subset $H \subseteq A^-$ with the following properties:

1. $1 \in H$;
2. H is a relatively complete subset of A ;
3. H is closed under multiplication;
4. for all $a \in H$, $a^2 = a$.

Lemma 7.1 *Let $\mathbf{A}$ be a Girard algebra and let H be a relatively complete Heyting subset of A . If we define $!_H a = \sup\{b \in H : b \leq a\}$, then $\langle \mathbf{A}, !_H \rangle$ is a girale. Conversely if $\mathbf{A}$ is a girale then $H = !A = \{!a : a \in A\}$ is a relatively complete Heyting subset of $\mathbf{A}$ and $!_H a = !a$.*

Proof We have to check that $!_H$ satisfies (G1)-(G4); (G1) is obvious since $H \subseteq A^-$ and $1 \in H$ and (G2) follows from the definition of $!_H$. Since clearly $a \leq b$ implies $!_H a \leq !_H b$ and $!_H a !_H a = !_H a$, from $!_H(a \wedge b) \leq !_H a, !_H b$ we get $!_H(a \wedge b) \leq !_H a !_H b$. For the converse, note that $!_H a \leq a \wedge 1$ and $!_H b \leq b \wedge 1$; since H is closed under multiplication we get

$$!_H a !_H b = (!_H a !_H b)(!_H a !_H b) \leq (a \wedge 1)(b \wedge 1) \leq a \wedge b \wedge 1.$$

This proves that $!_H a !_H b \leq !_H(a \wedge b)$ and hence (G3). Finally (G4) is obvious from the definition of $!_H$.

Since $1 \in !A$ by (G1), $!A \subseteq A^-$ by (G2), it is closed under products by (G3) and consists of idempotents by Lemma 6.2(3), we need only to show that

$$!a = \sup\{!b : !b \leq a\}.$$

Now if $b \leq !a$, then $!b \leq !a$ by Lemma 6.2(1); so $!a$ is an upper bound. Let $!b \leq c$ for all $!b \in !A$ such that $!b \leq a$; since $!a \leq a$ we get that $!a \leq c$, so $!a$ is the least upper bound. $\square$

Corollary 7.2 *Every complete Girard algebra is embeddable in (as a matter of fact, it is a reduct of) a girale.*

Proof In this case the set $H = \{a : a \leq 1 \text{ and } a^2 = a\}$ is a nonempty relatively complete Heyting subset of A and Lemma 7.1 applies. $\square$

So to prove that Girard algebras are exactly subreducts of girales it is enough to show that any Girard algebra can be embedded in a complete Girard algebra. Now we could be tempted to use the same embedding we used for Girard lattices; as a matter of fact it is not hard to show that if $\mathbf{A}$ is a commutative residuated lattice then

$\mathbf{A}$ is embeddable in the complete and commutative residuated lattice $D(\mathbf{A})$ (see [3] for details). However introducing an involutive De Morgan negation causes problems; these problems can of course be solved by constructing a different embedding using for instance the circle of ideas in [6], but we have a different and more direct embedding that does the job and we proceed to illustrate it.

Let $\mathbf{A}$ be a Girard algebra; we define a binary relation R on A by $(a, b) \in R$ if and only if $\sim b \not\geq a$; this relation is symmetric since $\sim$ is De Morgan. As for all binary relations there is a closure operator Q naturally associated to it; if $U \subseteq A$ we can define $Q(U) = \{a : (u, a) \in R \text{ for some } u \in U\}$. It is a standard exercise to prove that Q is a closure operator on A and hence the closed sets form a complete lattice with universe $C(\mathbf{A})$.

Lemma 7.3 *Let $\mathbf{A}$ be a girard algebra; then $\mathbf{C}(\mathbf{A}) = \langle C(\mathbf{A}), \vee, \wedge, \rightarrow, \cdot, \sim, \mathbf{1} \rangle$ is a complete Girard algebra upon defining for $X, Y \in C(\mathbf{A})$*

$$\begin{aligned} X \cdot Y &= Q(\{ab : a \in X, b \in Y\}) \\ \sim X &= \{b : \sim c \not\geq b \text{ implies } c \geq a \text{ for all } a \in X\} \\ X \rightarrow Y &= \sim(X \cdot \sim Y) \\ \mathbf{1} &= Q(\mathbf{1}). \end{aligned}$$

By our previous discussion to prove the lemma it is enough to show that the negation defined above is involutive and satisfies contraposition; this is a simple exercise, using the analogous properties of the negation in Girard algebras, and we leave it to the reader. The next lemma is more important.

Lemma 7.4 *Let $\mathbf{A}$ be a Girard algebra and for $a \in A$ let's denote by (a) the principal ideal generated by a . Then*

1. *for any $a \in A$, $(a) = Q(a) \in C(\mathbf{A})$;*
2. *the mapping $a \mapsto (a)$ is an embedding of $\mathbf{A}$ in $\mathbf{C}(\mathbf{A})$.*

Proof Observe that

$$Q(a) = \{b : c \not\geq b \text{ implies } c \not\geq a\}.$$

Suppose that $b \notin (a)$; then $b \not\leq a$ and then $b \notin Q(a)$. Conversely if $b \notin Q(a)$, then there exists a c with $\sim c \not\geq b$, and $\sim c \geq a$; hence $b \not\leq a$ and so $b \notin (a)$. This proves (1).

For (2) it is obvious that the mapping is a meet homomorphism. Let's show that

$$(a) \vee (b) = (a \vee b)$$

Observe that $(a) \vee (b) = Q((a) \cup (b))$. Since $(a) \cup (b) \subseteq (a \vee b)$ and the latter is closed, one inclusion is clear. Next observe that

$$(a) \vee (b) = \{c : \sim d \not\geq c \text{ implies } (\sim d \not\geq e \text{ for some } e \leq a \text{ or } \sim d \not\geq f \text{ for some } f \leq b)\}.$$

Let $c \notin (a] \vee (b]$. Then there exists a d with $\sim d \not\geq c$ but $\sim d \geq a$ and $\sim d \geq b$. Hence $\sim d \geq a \vee b$ and so, since $\sim d \not\geq c$, $c \notin (a \vee b]$.

A similar argument shows that

$$(a] \cdot (b] = Q(\{uv : u \leq a, v \leq b\}) = (ab].$$

Next we check that

$$(\sim a] = \sim(a] = \{b : c \not\geq b \text{ implies } c \not\leq a\}.$$

Suppose that $b \notin \sim(a]$. Then there exists a c with $\sim c \not\geq b$ and $c \leq a$. Thus $b \not\leq \sim a$, otherwise $b \leq \sim a \leq \sim c$. Conversely if $b \not\leq \sim a$, then $\sim a \not\geq b$ and thus $b \notin \sim(a]$.

Since $\rightarrow$ is definable in both cases by the negation the mapping is a homomorphism and it is obviously injective. $\square$

Corollary 7.5 *Every (bounded) Girard algebra is embeddable in a (bounded) girale; hence the variety of (bounded) Girard algebras is exactly the class of subreducts of (bounded) girates. Therefore LL is a conservative extension of MALL.*

8 Congruences

Congruences in commutative residuated lattices are well known; since the variety CRL of commutative residuated lattices is ideal-determined in the sense of [5] the congruences are completely determined by certain subsets of the universe that we call **U-ideals**. In case of CRL these subsets have a particularly transparent description; if $\mathbf{A} \in \text{CRL}$ a **filter** of $\mathbf{A}$ is a subset $F \subseteq A$ such that

- F is a lattice filter;
- $1 \in F$;
- if $a, a \rightarrow b \in F$, then $b \in F$.

Since the intersection of any family of filters is clearly a filter, there is a closure operator on A in which the closed sets are exactly the filters; the operator is easily shown to be algebraic, so the filters of $\mathbf{A}$ form an algebraic lattice $\text{Fil}(\mathbf{A})$. The following fact was observed in [3] and has been rediscovered many times since.

Theorem 8.1 *If $\mathbf{A} \in \text{CRL}$ then $\text{Con}(\mathbf{A})$ and $\text{Fil}(\mathbf{A})$ are isomorphic through the mappings $\theta \mapsto 1/\theta$ and $F \mapsto \theta_F = \{(a, b) : a \rightarrow b, b \rightarrow a \in F\}$.*

Now it is evident that a congruence of a Girard algebra is a congruence of its underlying commutative residuated lattice structure, so Theorem 8.1 specializes easily to Girard algebras. What about girates? They are still ideal-determined, so the congruences are totally determined by the U-ideals. However the operation $!$ is a *compatible operation* in the sense of [3]; this in turn implies that the U-ideals of a girale $\mathbf{A}$ are just the filters of its underlying Girard algebra, that are closed under $!$. We will name these subsets *filters* as well and let the context clear the meaning. In any case Theorem 8.1 holds: if $\mathbf{A}$ is a girale its congruence lattice is isomorphic with the lattice of filters. Moreover we get the following useful description:

Lemma 8.2 *Let $\mathbf{A}$ be a girale, let $X \subseteq A$ and let $\text{Fil}_{\mathbf{A}}(X)$ the filter generated by X ; then*

$$\text{Fil}_{\mathbf{A}}(X) = \{a : !b_1 \dots !b_n \leq a \text{ for some } b_1, \dots, b_n \in X\}.$$

Proof Let F be the set described by the right hand side of the equality. Then F is a lattice filter, since it is the union of a directed family of lattice filters; moreover if $a \in F$, then there are $b_1, \dots, b_n \in X$ such that

$$!(b_1 \wedge \dots \wedge b_n) = !b_1 \dots !b_n \leq a.$$

This implies immediately that $!a \in F$, so F is a filter which contains X . If G is another filter containing X then $F \subseteq G$, so $F = \text{Fil}_{\mathbf{A}}(X)$. $\square$

Lemma 8.2 has obvious consequences; let $[a]$ denote the principal filter generated by a .

Corollary 8.3 *Let $\mathbf{A}$ be a girale; then for any $a \in A$, $\text{Fil}_{\mathbf{A}}(a) = [!a]$. In other words every principal filter of $\mathbf{A}$ is principal as a lattice filter.*

In [3] (Theorem 3.8) it has been shown that a variety $\mathbf{V}$ of commutative residuated semilattice with normal operators³ such that every principal filter of any algebra $\mathbf{A} \in \mathbf{V}$ is principal as a semilattice filter, must have a (commutative, regular) TD-term in the sense of [13].

Now girates are lattices and $!$ is a normal operator; moreover the fact that we have a De Morgan negation implies that the join is definable with meet and negation (which is in turn definable using $\rightarrow$ and $\mathbf{0}$). It follows that girates can be seen as commutative residuated semilattices and the above result applies. If $\mathbf{A}$ is an algebra, we denote by $\text{Cg}_{\mathbf{A}}(a, b)$ the principal congruence generated by (a, b) in $\mathbf{A}$. We have:

Corollary 8.4 *The variety $\mathbf{Gi}$ of girates has a TD-term and thus it has equationally definable principal congruences. For any $\mathbf{A} \in \mathbf{Gi}$ and $a, b, c, d \in A$*

$$(c, d) \in \text{Cg}_{\mathbf{A}}(a, b) \quad \text{if and only if} \quad !(a \leftrightarrow b) \leq c \leftrightarrow d.$$

Through the general theory of algebraizable logics we get that LL has the deduction theorem: $\Gamma, p \vdash_{\text{LL}} q$ if and only if $\Gamma \vdash_{\text{LL}} !p \rightarrow q$. But this is of course well known [8].

From Lemma 7.1 we know that $!A$ is a relatively complete Heyting subset of $\mathbf{A}$; with some adjustment it can be made into a Heyting algebra $!\mathbf{A}$ which is deeply connected with $\mathbf{A}$.

Theorem 8.5 *For any girale $\mathbf{A}$, $!\mathbf{A} = (!A, \vee, \wedge, \rightarrow, \perp, 1)$ is a Heyting algebra, where for $u, v \in !A$*

$$u \wedge !v = u \wedge v \quad u \rightarrow !v = !(u \rightarrow v).$$

³ If $\mathbf{A}$ is a residuated semilattice, a normal operator is a unary operation f on A such that $f(1) = 1$ and $f(a \wedge 1) = f(a) \wedge 1$ for all $a \in A$; moreover f must preserve the lattice ordering and for all $a, b \in A$, $f(a)f(b) \leq f(ab)$.

Moreover $\text{Con}(\mathbf{A}) \cong \text{Con}(!\mathbf{A})$.

Proof We observe first that $!$ is a *conucleus* in the sense of [24]; then we apply Lemma 3.1 in [24] to conclude that $!\mathbf{A}$ is a commutative residuated lattice. Since $!1 = 1$, it is also integral and by Lemma 6.2(3) every element of $!\mathbf{A}$ is idempotent. This is enough to deduce that $!\mathbf{A}$ is a Heyting algebra.

Next we show that the mappings

$$H \longmapsto H \cap !A \quad G \longmapsto \text{Fil}_{!\mathbf{A}}(G)$$

induce a lattice isomorphism between the filter lattice of $\mathbf{A}$ and the filter lattice of $!\mathbf{A}$. Since they both clearly preserve the ordering we need only check that they are well defined and their composition is the identity on the respective domains.

That for any filter H of $\mathbf{A}$, $H \cap !A$ is a filter of $!\mathbf{A}$, it is a consequence of Lemma 6.2(4). Now let G be a filter of $!\mathbf{A}$ and let $H = \text{Fil}_{\mathbf{A}}(G)$. We will show that $H \cap !A = G$. Clearly $G \subseteq H \cap !A$. If $!a \in H$, then by Lemma 8.2 there are $!b_1, \dots, !b_n \in G$ such that

$$!b_1 \dots !b_n \leq !a.$$

But since

$$!b_1 \dots !b_n = !(b_1 \wedge \dots \wedge b_n) = !b_1 \wedge ! \dots \wedge !b_n$$

by the usual description of filters in a Heyting algebras we get that $!_F a \in G$.

On the other hand if H is a filter of $\mathbf{A}$, it is obvious that $\text{Fil}_{!\mathbf{A}}(H \cap !A) = H$, since $!a \leq a$. $\square$

Since simple and subdirectly irreducible Heyting algebras are well-known we get immediately:

Corollary 8.6 *Let $\mathbf{A}$ be any girale:*

1. $\mathbf{A}$ is simple if and only if $|\mathbf{A}| = 2$;
2. $\mathbf{A}$ is subdirectly irreducible if and only if there is an $a \in A \setminus \{1\}$ such that $!b \leq !a$ for all $b \in A$.

Let us define an operator Heyt on a class $\mathbf{K}$ of girales by

$$\text{Heyt}(\mathbf{K}) = \{!\mathbf{A} : \mathbf{A} \in \mathbf{K}\}.$$

Theorem 8.7 *For any variety $\mathbf{V}$ of girales, $\text{Heyt}(\mathbf{V})$ is a variety of (pointed) Heyting algebras.*

Proof We need only to show that $\text{Heyt}(\mathbf{V})$ is closed under **H**, **S** and **P**. Let $!\mathbf{A} \in \text{Heyt}(\mathbf{V})$ and let $G \in \text{Fil}(!\mathbf{A})$. Let $F = \{b \in A : !a \leq b \text{ for some } !a \in G\}$; then F is a filter of $\mathbf{A}$ and $F \cap !A = G$. So $!(\mathbf{A}/F) \cong !\mathbf{A}/G$ and $!\mathbf{A}/G \in \text{Heyt}(\mathbf{V})$. That $\text{Heyt}(\mathbf{V})$ is closed under direct products is obvious so let $!\mathbf{A} \in \text{Heyt}(\mathbf{V})$ and let $\mathbf{C}$ be a subalgebra of $!\mathbf{A}$ and let $\mathbf{B}$ the subalgebra of $\mathbf{A}$ generated by C . Clearly $C \subseteq !\mathbf{B}$; conversely, as

C generates **B** in **A**, every element of **!B** is $!t(c_1, \dots, c_n)$ for some n -ary term of **A** and $c_1, \dots, c_n \in C$. Now an induction on the complexity of $t(x_1, \dots, x_n)$ shows that if $c_1, \dots, c_n \in C$ then $!t(c_1, \dots, c_n) \in !C$. The only nontrivial part is to show that if $c \in C$, then $!\sim!c \in C$.

Observe that from $\sim!c!c \leq \sim 1$ and $!\sim!c \leq \sim!c$ we get $!\sim!c!c \leq \sim 1$ and hence $!\sim!c!c \leq !\sim 1$.

By residuation $!\sim!c \leq !c \rightarrow !\sim 1$ and hence $!\sim!c \leq !(c \rightarrow !\sim 1) = !c \rightarrow !\sim 1$. On the other hand

$$\sim!c = !c \rightarrow \sim 1 \geq !c \rightarrow !\sim 1,$$

implying

$$!\sim!c \geq !(c \rightarrow !\sim 1) = !c \rightarrow !\sim 1.$$

In conclusion $!\sim!c = !c \rightarrow !\sim 1 \in C$. □

The mapping $V \mapsto \text{Heyt}(V)$ is a join homomorphism from the lattice of subvarieties of girales to the lattice of subvarieties of Heyting algebras:

$$\begin{aligned} \text{Heyt}(V \vee V') &= \text{Heyt}(\mathbf{V}(V \cup V')) = \mathbf{V}(\text{Heyt}(V \cup V')) \\ &= \mathbf{V}(\text{Heyt}(V) \cup \text{Heyt}(V')) = \text{Heyt}(V) \vee \text{Heyt}(V'). \end{aligned}$$

However it is not a meet homomorphism, as we shall see later.

Thus the lattice of varieties of girales can be partitioned into equivalence classes that are also join semilattices; information on these classes can be recovered from the varieties of Heyting algebras that are their “natural” representatives. Such pieces of information can be glued together to get a clearer picture of the whole lattice of varieties of girales.

Let's call a girale **A** a **Boolean girale** if $\text{Heyt}(\mathbf{A})$ is a Boolean algebra; a variety **V** of girales is **Boolean** if every algebra in **V** is Boolean. We will give a recipe to construct a family of simple Boolean girales of unbounded cardinality. By Jónsson Lemma [22], any two of them generate distinct Boolean varieties of girales.

Let G_n be the height three lattice with n atoms. Let $\perp$ and $\top$ be the bottom and the top of the lattice and let $1, 0$ be two distinct atoms. Define $\sim$ on G_n by setting $\sim\perp = \top$, $\sim\top = \perp$, $\sim 1 = \perp$, $\sim 0 = 1$ and $\sim a = a$ for any other $a \in G_n$. Define $\cdot$ on G_n by

$$\begin{aligned} \perp \cdot a &= a \cdot \perp = \perp \\ 1 \cdot a &= a \cdot 1 = a \\ a \cdot a &= \perp & a \notin \{0, 1, \perp, \top\} \\ a \cdot b &= \top & \text{otherwise} \end{aligned}$$

Finally define $!1 = !\top = 1$ and $!a = \perp$ otherwise. Some calculations show that each G_n is a Boolean girale; the subdirectly irreducible algebras in $\mathbf{V}(G_n)$ are exactly the G_m , $1 \leq m \leq n$.

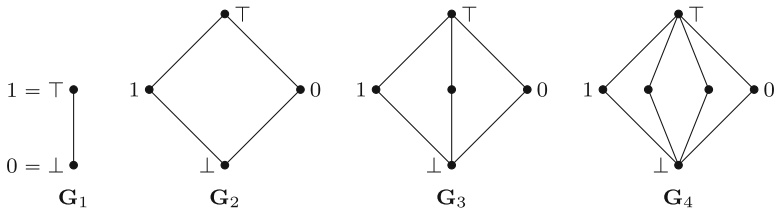


Fig. 1 Boolean girales

The algebra $\mathbf{G}_1$ in Fig. 1 is the two element Boolean algebra and hence $\text{Heyt}(\mathbf{V}(\mathbf{G}_1)) = \mathbf{B}$, the variety of Boolean algebras. On the other hand, since both $\mathbf{G}_1$ and $\mathbf{G}_2$ are finite simple algebras with no proper subalgebras both $\mathbf{V}(\mathbf{B})$ and $\mathbf{V}(\mathbf{G}_2)$ are atoms in the lattice of varieties of girales and their intersection is the trivial variety. However $\text{Heyt}(\mathbf{V}(\mathbf{G}_2)) = \mathbf{B}$ as well, so $\text{Heyt}(\mathbf{V}(\mathbf{G}_1)) \cap \text{Heyt}(\mathbf{V}(\mathbf{G}_2)) = \mathbf{B}$. This shows that the map $\mathbf{V} \mapsto \text{Heyt}(\mathbf{V})$ is not a meet homomorphism.

9 Other fragments and expansions of LL

9.1 Fragments

Removing **1** from the language is equivalent to consider a logic without a defined notion of *truth*. This was actually the way in which Relevance Logic was originally presented and it is yet another point of contact; not surprisingly what happens here is very similar to what happens in Relevance Logic.

We will describe in more details the case of the **1**-less reduct of MALL and leave all the others to the interested reader. Let LR be the system axiomatized by the axioms (HL1)–(HL7), (HL12)–(HL19), (MP) and (Adj). The system is algebraizable with defining equation $p \wedge (p \rightarrow p) = p \rightarrow p$ and equivalence formulas $\Delta = \{p \rightarrow q, q \rightarrow p\}$. The Lindenbaum algebra of LR is easily seen to be a bounded lattice with involution and with an implication satisfying (2.2)–(2.5). Unfortunately the variety $\mathbf{V}$ of such algebras is not the equivalent algebraic semantics of LR, since the natural ordering of the underlying lattice does not model adequately the residuation in LR. It turns out that the variety of LR-algebras, i.e. the subvariety of $\mathbf{V}$ axiomatized by the equation

$$((x \rightarrow x) \wedge (y \rightarrow y)) \rightarrow z \leq z,$$

is the equivalent algebraic semantics for LR.

With an argument similar to the one for Relevance Logic one can show that LR does not have equationally definable principal congruences. However, if we add to LR the *mingale axiom*

$$p \rightarrow (p \rightarrow p),$$

then the resulting equivalent algebraic semantics has equationally definable principal congruences and thus the system has the deduction theorem. This displays once more the strong connections between Linear Logic and Relevance Logic.

The implicational fragment of LL gives rise to a well known deductive system, BCI-logic, which is not algebraizable [11]. In [12], Blok and Pigozzi noted also that the $\{\rightarrow, \cdot\}$ -fragment of LL is not algebraizable. We describe briefly its logical matrices: let $\langle A, \rightarrow, \cdot, \leq \rangle$ be a commutative partially ordered residuated semigroup. A *filter* of A is an order filter of A closed under multiplication. One sees easily that a reduced matrix for the $\{\rightarrow, \cdot\}$ -fragment of LL is $\langle A, \nabla_A \rangle$ where A is a commutative p.o. residuated semigroup and ∇_A is the filter generated by the set $\{a \rightarrow a : a \in A\}$. Hence the reduced matrix semantics is the class of *reduced filtered commutative residuated partially ordered semigroups*. The reason why this class cannot be replaced by a class of proper algebras lies in the fact that the partial order cannot be recovered from the operations. By the same argument one sees that the multiplicative fragment of LL is not algebraizable.

Finally we observe that algebraizable fragment whose language contains ! has the deduction theorem and gives rise to an equivalent algebraic semantics having equationally definable principal congruences.

9.2 Intuitionistic Linear Logic

There are at least two versions of intuitionistic linear logic. The first one is ILL (see [20]) which is obtained from classical linear logic in the same way Gentzen's system LJ (intuitionistic logic) is obtained from LK: a left side of a sequent can contain at most one formula. So we are forced to drop all connectives and constants whose rules do not obey to this restriction, i.e. $\neg, \perp, \wp$. ILL can be seen as a deductive system axiomatized Hilbert-style by (HL1–HL3), (HL6–HL9), (HL12–HL24) with (MP), (Adj) and (Nec).

It is sometimes convenient, to get a more suitable comparison with classical linear logic, to consider ILN [1], i.e. *intuitionistic Linear Logic with negation*. This is done by adding two “ad hoc” rules for $\sim$ in the sequent calculus. On the Hilbert-style side we just add (HL5), (HL10), (HL11) (one of course has to define $\perp$ as ~ 1). Finally, if we drop (HL20–HL24) and (Nec) from ILL or ILN we get the exponential-free fragments.

It is clear that all these systems are fragments of LL with respect to suitable languages. Since the connectives appearing in the congruence formulas and defining equation for LL belong to all these languages, we can conclude at once (by Corollary 2.12 in [11]) that all the systems are algebraizable (and most of them in fact strongly algebraizable).

9.3 Noncommutative Linear Logics

If in the sequent formulation of Linear Logic we delete the *exchange rule*, we get *noncommutative Linear Logic*. It is not hard (only boring) to work out a Hilbert-style axiomatization of non commutative Linear Logic. In this framework one has to deal with two implications and two negations and the product $\cdot$ is no longer commutative. Of course also in this case we have interesting fragments and in particular one can get *noncommutative intuitionistic linear logic* [2]. Noncommutative Linear Logics are algebraizable (the proof is similar to the one for Linear Logic, only more work has to

be done to take care of the doubling of implication and negation) and the equivalent algebraic semantics are varieties of FL-algebras with further operations.

10 Conclusions

Linear Logic as a deductive system has not been investigated much and its algebraic counterparts, i.e. the varieties of Girard algebras and g-irales, even less. But things are changing; a very interesting recent contribution is in [18], where the author obtained significant information about interpolation in Linear Logic using some of the results in this note.

This is, in my opinion, a compelling motivation for starting a vigorous investigation of the algebraic semantics of Linear Logic. Of course many facts about Girard algebras can be recovered using the general theory of commutative residuated lattices with an involutive negation (see for instance [19]), but the variety of g-irales can be interesting also from a different viewpoint. In the past years there have been a flurry of papers in which a modality of some kind is superimposed to residuated lattices, mostly trying to generalize the historic work on monadic algebras done by Halmos [21]; the most general attempt is in O. Tuyt's Ph.D. Thesis [27] which also constitutes an excellent survey on the subject.

Now while these results are certainly interesting, sometimes, at least in my opinion, the motivation for adding a modal operator is not totally convincing; g-irales on the other hand, constitute a perfect example of a residuated lattice in which the introduction of an S_4 modality is totally justified and natural and perhaps attacking them with the usual arsenal of algebraic logic might be worthwhile. For instance studying the various versions of structural completeness in sub(quasi)varieties of g-irales (see [4]) might give some relevant information. We hope to be able to see (and do) more along these lines in the future.

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